

## KUNGRAD, Uzbekistan

WMO: 381490

Lat: 43.083N

Lon: 58.933E

Elev: 60

StdP: 100.61

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -20.0      | -16.9 | -23.8                         | 0.4 | -19.5 | -20.7 | 0.6 | -16.5 | 14.7                  | -2.8 | 11.8 | -1.2 | 2.2                   | 10   | 0.553 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 14.7                   | 39.3            | 21.5 | 37.7 | 20.9 | 36.2 | 20.6 | 22.7                | 35.8 | 21.9 | 35.4 | 21.2 | 34.4 | 3.7                  | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.3                            | 13.3 | 28.7 | 17.2 | 12.4 | 28.2 | 16.3 | 11.7 | 27.8 | 67.0          | 36.1 | 64.3 | 35.4 | 61.8 | 34.5 | 26.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 13.9          | 10.4 | 8.5 | -22.0                      | 41.9 | 4.4                | 1.7 | -25.2                                              | 43.1 | -27.8      | 44.1 | -30.2      | 45.1 | -33.5      | 46.4 |    |    |
| (5)               |      |     | -22.3                      | 24.4 | 4.3                | 1.0 | -25.4                                              | 25.1 | -27.9      | 25.7 | -30.3      | 26.2 | -33.5      | 26.9 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 12.3   | -4.4  | -2.2  | 5.7   | 14.3  | 21.6  | 26.3  | 28.1  | 26.0  | 19.2  | 11.4  | 3.3   | -2.4  |  |  |
|                                                                        | DBStd       | 12.48  | 5.94  | 6.04  | 5.56  | 5.32  | 3.86  | 3.25  | 2.86  | 3.39  | 3.90  | 4.79  | 5.38  | 5.09  |  |  |
|                                                                        | HDD10.0     | 1597   | 447   | 343   | 154   | 20    | 0     | 0     | 0     | 0     | 1     | 41    | 208   | 384   |  |  |
|                                                                        | HDD18.3     | 3177   | 705   | 576   | 391   | 142   | 16    | 1     | 0     | 1     | 35    | 218   | 451   | 642   |  |  |
|                                                                        | CDD10.0     | 2444   | 0     | 1     | 22    | 149   | 359   | 488   | 559   | 497   | 277   | 85    | 7     | 0     |  |  |
|                                                                        | CDD18.3     | 981    | 0     | 0     | 1     | 21    | 116   | 239   | 301   | 239   | 62    | 4     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 13373  | 0     | 1     | 14    | 281   | 1444  | 3299  | 4239  | 3213  | 805   | 77    | 1     | 0     |  |  |
|                                                                        | CDH26.7     | 7358   | 0     | 0     | 3     | 95    | 636   | 1894  | 2547  | 1866  | 307   | 12    | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 3.6    | 3.3   | 3.8   | 4.3   | 4.4   | 4.2   | 3.8   | 3.3   | 3.2   | 3.1   | 3.1   | 3.1   | 3.3   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 114    | 9     | 9     | 16    | 17    | 16    | 7     | 3     | 2     | 4     | 6     | 13    | 10    |  |  |
|                                                                        | PrecMax     | 239    | 26    | 49    | 48    | 53    | 77    | 47    | 13    | 13    | 17    | 32    | 56    | 30    |  |  |
|                                                                        | PrecMin     | 36     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | PrecStd     | 49     | 6     | 10    | 12    | 13    | 17    | 9     | 4     | 3     | 5     | 8     | 14    | 8     |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 10.0  | 15.4  | 25.1  | 32.6  | 36.6  | 40.7  | 42.1  | 41.0  | 35.9  | 28.3  | 20.2  | 12.1  |  |  |
|                                                                        |             | MCWB   | 6.3   | 9.8   | 14.8  | 18.5  | 20.1  | 21.8  | 22.3  | 22.2  | 20.4  | 16.6  | 13.3  | 7.6   |  |  |
|                                                                        | 2%          | DB     | 6.5   | 10.9  | 20.7  | 29.5  | 34.2  | 38.5  | 39.6  | 38.6  | 32.2  | 25.6  | 16.4  | 8.4   |  |  |
|                                                                        |             | MCWB   | 3.7   | 6.9   | 13.1  | 17.6  | 19.5  | 21.1  | 21.6  | 21.3  | 19.1  | 15.3  | 10.5  | 5.0   |  |  |
|                                                                        | 5%          | DB     | 4.4   | 7.9   | 17.5  | 26.6  | 32.3  | 36.6  | 37.9  | 36.8  | 30.1  | 23.3  | 13.9  | 6.1   |  |  |
|                                                                        |             | MCWB   | 2.1   | 4.5   | 11.2  | 16.2  | 18.8  | 20.6  | 21.0  | 20.9  | 18.0  | 14.4  | 8.9   | 3.5   |  |  |
|                                                                        | 10%         | DB     | 2.7   | 5.3   | 14.7  | 23.8  | 30.3  | 35.0  | 36.3  | 35.0  | 28.0  | 20.6  | 11.3  | 4.3   |  |  |
|                                                                        |             | MCWB   | 0.7   | 2.6   | 9.5   | 14.8  | 18.1  | 20.2  | 20.6  | 20.2  | 17.3  | 12.9  | 7.5   | 2.0   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 6.6   | 10.3  | 15.6  | 19.8  | 22.0  | 23.6  | 23.7  | 23.6  | 21.1  | 17.6  | 13.9  | 7.8   |  |  |
|                                                                        |             | MCDB   | 9.5   | 15.2  | 24.5  | 29.5  | 33.1  | 35.7  | 36.8  | 36.6  | 32.8  | 26.8  | 19.5  | 11.2  |  |  |
|                                                                        | 2%          | WB     | 3.9   | 6.9   | 13.1  | 18.1  | 20.7  | 22.3  | 22.6  | 22.3  | 19.7  | 16.0  | 11.0  | 5.3   |  |  |
|                                                                        |             | MCDB   | 6.2   | 10.9  | 20.2  | 28.0  | 31.8  | 35.5  | 36.1  | 35.9  | 31.2  | 24.4  | 15.7  | 8.1   |  |  |
|                                                                        | 5%          | WB     | 2.3   | 4.6   | 11.4  | 16.7  | 19.6  | 21.5  | 21.9  | 21.5  | 18.5  | 14.6  | 9.2   | 3.6   |  |  |
|                                                                        |             | MCDB   | 4.3   | 7.7   | 17.4  | 25.9  | 30.5  | 34.7  | 35.6  | 35.0  | 29.2  | 22.6  | 13.2  | 6.0   |  |  |
|                                                                        | 10%         | WB     | 0.9   | 2.7   | 9.6   | 15.2  | 18.6  | 20.7  | 21.2  | 20.7  | 17.5  | 13.2  | 7.7   | 2.1   |  |  |
|                                                                        |             | MCDB   | 2.6   | 5.1   | 14.5  | 23.1  | 29.0  | 33.3  | 34.5  | 33.8  | 27.3  | 20.1  | 11.3  | 4.2   |  |  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 8.3   | 10.0  | 11.7  | 12.7  | 13.5  | 14.6  | 14.7  | 15.7  | 15.2  | 14.2  | 10.7  | 8.4   |  |  |
|                                                                        | 5% DB       | MCDBR  | 9.5   | 13.6  | 16.0  | 16.0  | 16.2  | 17.1  | 16.9  | 18.3  | 17.6  | 17.6  | 14.8  | 11.3  |  |  |
|                                                                        |             | MCWBR  | 7.5   | 10.4  | 10.1  | 8.1   | 6.9   | 6.1   | 5.4   | 6.6   | 8.1   | 9.8   | 10.1  | 8.9   |  |  |
|                                                                        | 5% WB       | MCDBR  | 8.9   | 13.0  | 15.6  | 15.3  | 15.1  | 15.6  | 15.4  | 17.2  | 17.5  | 16.3  | 13.6  | 10.6  |  |  |
|                                                                        |             | MCWBR  | 7.1   | 10.0  | 10.1  | 8.5   | 7.6   | 6.6   | 5.9   | 7.0   | 8.6   | 9.3   | 9.6   | 8.5   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.379 | 0.405 | 0.470 | 0.525 | 0.529 | 0.513 | 0.497 | 0.472 | 0.445 | 0.450 | 0.411 | 0.382 |  |  |
|                                                                        | taud        |        | 2.072 | 2.019 | 1.909 | 1.814 | 1.851 | 1.920 | 1.988 | 2.026 | 2.039 | 1.971 | 2.036 | 2.064 |  |  |
|                                                                        | Ebn at Noon |        | 705   | 758   | 755   | 748   | 760   | 774   | 782   | 786   | 773   | 696   | 657   | 655   |  |  |
|                                                                        | Edh at Noon |        | 108   | 135   | 171   | 203   | 201   | 188   | 174   | 162   | 148   | 140   | 112   | 100   |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.50  | 2.51  | 3.92  | 5.38  | 6.63  | 7.43  | 7.16  | 6.57  | 5.18  | 3.41  | 2.02  | 1.36  |  |  |
|                                                                        | RadStd      |        | 0.17  | 0.30  | 0.32  | 0.37  | 0.37  | 0.33  | 0.26  | 0.24  | 0.21  | 0.20  | 0.24  | 0.18  |  |  |

## Historical Trends

|      |                        | DBAvg |     | Heating |        | Cooling |       | Degree-Days |         |         |         |         |
|------|------------------------|-------|-----|---------|--------|---------|-------|-------------|---------|---------|---------|---------|
|      |                        |       |     | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP       | HDD10.0 | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                        | (d)   | (e) | (f)     | (g)    | (h)     | (i)   | (j)         | (k)     | (l)     | (m)     |         |
| (46) | Station Only           | N/A   | N/A | N/A     | +1.16  | N/A     | -0.71 | N/A         | N/A     | N/A     | +98     | (46)    |
| (47) | Regional (4 neighbors) | N/A   | N/A | N/A     | +0.84  | N/A     | -0.63 | N/A         | N/A     | N/A     | +63     | (47)    |

Nomenclature: See separate page